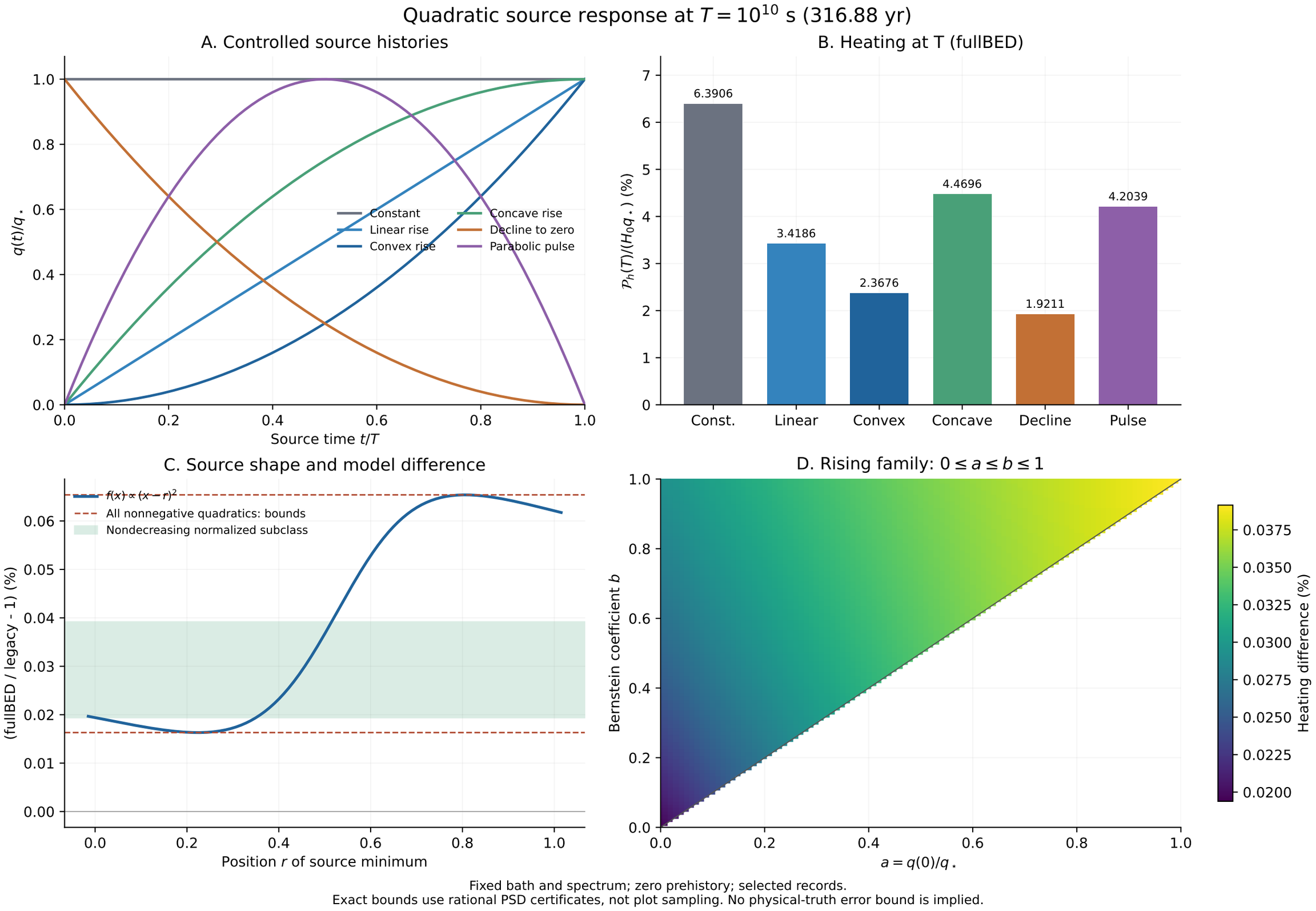
Quadratic source response at $T = 10^{10}$ s (316.88 yr)
A. Controlled source histories
$q(t)/q_\star$
Source time t/T
Constant
Linear rise
Convex rise
Concave rise
Decline to zero
Parabolic pulse
B. Heating at T (fullBED)
$\mathcal{P}_h(T)/(H_0 q_\star)$ (%)
6.3906
3.4186
2.3676
4.4696
1.9211
4.2039
Const.
Linear
Convex
Concave
Decline
Pulse
C. Source shape and model difference
$f(x) \propto (x - r)^2$
All nonnegative quadratics: bounds
Nondecreasing normalized subclass
(fullBED / legacy - 1) (%)
Position r of source minimum
D. Rising family: $0 \le a \le b \le 1$
Bernstein coefficient b
$a = q(0)/q_\star$
Heating difference (%)
Fixed bath and spectrum; zero prehistory; selected records.
Exact bounds use rational PSD certificates, not plot sampling. No physical-truth error bound is implied.